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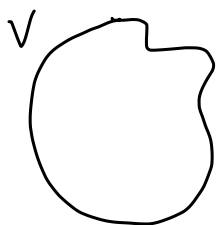
$$u(\vec{x}, t)$$

$$\cancel{\frac{\delta u}{\delta t}} + \nabla \cdot \vec{I} = f$$

VOLUME CONDUCTOR THEORY $\longrightarrow$ CONSERVED ? q ELECTRICAL CHARGE

$$u \rightarrow \left[\frac{C}{m^3} \right]$$

CONSERVATION OF CHARGE $\longrightarrow \frac{\delta u}{\delta t} = 0$

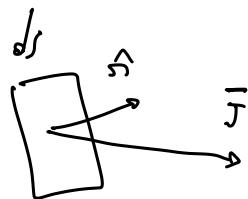


$$Q_{TOT} = \text{const} \longrightarrow \forall V$$

$$Q_{TOT} = \int_V u \, dV = \text{const} \quad \frac{\delta Q_{TOT}}{\delta t} = 0$$

$$\int_V \frac{\delta u}{\delta t} \, dV = 0 \quad \forall V \implies \frac{\delta u}{\delta t} = 0$$

$$\vec{I} = \vec{J} \quad \text{CURRENT DENSITY} \quad \left[\frac{mA}{cm^2} \right]$$



$$\vec{J} \cdot \hat{n} \, dS \longrightarrow \text{CURRENT FLOWS THROUGH } dS \quad [mA]$$

$$f \rightarrow I_V(\bar{x}, t) \rightarrow \begin{array}{c} \text{CURRENT} \\ \text{SOURCE} \\ \text{DENSITY} \end{array} \quad \left[\frac{\text{mA}}{\text{m}^3} \right]$$

$$\nabla \cdot \bar{J} = f = I_V$$

$$\bar{J} = \underset{\substack{\uparrow \text{ELECTRIC FIELD} \\ \downarrow \text{CONDUCTIVITY}}}{\sigma} \bar{E} \quad \bar{E} = -\nabla \phi$$

$$\bar{J} = -\sigma \nabla \phi \quad \begin{array}{l} \bar{J}(\bar{x}, t) \\ \phi(\bar{x}, t) \end{array}$$

$$\sigma(\bar{x})$$

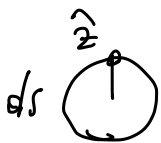
$I_V \neq 0$
in EXITABLE
TISSUES

$$\nabla \cdot (-\sigma \nabla \phi) = I_V$$

$$\nabla \cdot (\sigma(x) \nabla \phi(\bar{x}, t)) = I_V(x, t)$$

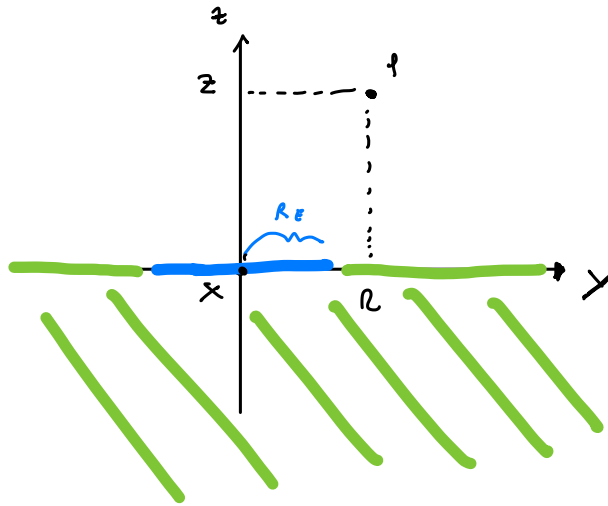
σ depends on x !

ELECTRODE DISK EXAMPLE

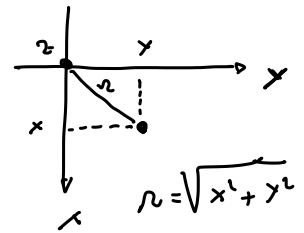


$$dI = \bar{J} \cdot \hat{z} \, dS = J_z \, dS \quad J_z(R)$$

$$I_s = \int_0^R J_z \, dS$$



INSULATION
ELECTRODE $V_S = 2V$



$$\phi, (n, z)$$

INSULATION CONDITION $\rightarrow \begin{cases} z=0 \\ n > R_E \end{cases}$

$$\begin{aligned} dI &= \vec{J} \cdot \hat{z} dS = \\ &= J_z dS = \phi \end{aligned}$$

$$\vec{J} = -\sigma \nabla \phi = -\sigma \left(\frac{\partial \phi}{\partial x} \hat{x} + \frac{\partial \phi}{\partial y} \hat{y} + \frac{\partial \phi}{\partial z} \hat{z} \right)$$

$$J_z = -\sigma \frac{\partial \phi}{\partial z}$$

$$-\sigma \frac{\partial \phi}{\partial z} dS = \phi \Rightarrow \frac{\partial \phi}{\partial z} = \phi \quad \begin{matrix} z=0 \\ n > R_E \end{matrix} \quad \begin{matrix} \text{ZERO} \\ \text{FLUX} \\ \text{CONDITION} \end{matrix}$$